

Electricity Price Forecasting

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**A dissertation preparation for the degree of Master of Science,
Specialisation Area of Information and Knowledge Systems**

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Abstract

This dissertation explores the application of advanced machine learning techniques to enhance electricity price forecasting in deregulated markets. As electricity markets evolve due to deregulation and increased competition, accurate price forecasting becomes critical for stakeholders, including energy producers, grid operators, and traders. Traditional forecasting models often struggle to capture the complex, non-linear dynamics of electricity prices influenced by various factors such as weather patterns, demand fluctuations, supply-side uncertainties, and regulatory changes.

This research employs a CRISP-DM methodology approach, utilizing machine learning algorithms such as Long Short-Term Memory (LSTM), Random Forest, and Gradient Boosting to improve both short-term and long-term price predictions. The methodology encompasses data collection and preprocessing, model development, feature engineering, and validation against real-world data from markets like the European Energy Exchange (EEX) and the New York Independent System Operator (NYISO).

Key findings indicate that integrating multiple data sources, including real-time weather data and demand forecasts, significantly enhances the accuracy and reliability of forecasting models. The results of the analyzed articles demonstrate that advanced machine learning techniques outperform traditional models, providing robust forecasts that can guide investment strategies and risk management in the evolving landscape of electricity markets. This work contributes to the ongoing discourse on optimizing energy trading and ensuring system stability through improved forecasting methodologies.

Keywords: Electricity Price Forecasting, Machine Learning, Deregulated Markets, Forecasting Models, Data Analysis

Resumo

Esta dissertação explora a aplicação de técnicas avançadas de machine learning para melhorar a previsão do preço da eletricidade em mercados desregularizados. À medida que os mercados de electricidade evoluem devido à desregulamentação e ao aumento da concorrência, a previsão precisa dos preços torna-se crítica para as partes interessadas, incluindo produtores de energia, operadores de rede e comerciantes. Os modelos de previsão tradicionais têm muitas vezes dificuldade em captar a dinâmica complexa e não linear dos preços da electricidade influenciada por vários factores, tais como padrões meteorológicos, flutuações da procura, incertezas do lado da oferta e alterações regulamentares.

Esta pesquisa emprega uma abordagem CRISP-DM, utilizando algoritmos de machine learning, como Long Short-Term Memory (LSTM), Random Forest e Gradient Boosting para melhorar as previsões de preços de curto e longo prazo. A metodologia abrange recolha e pré-processamento de dados, desenvolvimento de modelos e validação em relação a dados do mundo real de mercados como a Bolsa Europeia de Energia (EEX) e o Operador de Sistema Independente de Nova York (NYISO).

As principais conclusões indicam que a integração de múltiplas fontes de dados, incluindo dados meteorológicos em tempo real e previsões de procura, aumenta significativamente a precisão e a fiabilidade dos modelos de previsão. Os resultados dos artigos analisados demonstram que as técnicas avançadas de machine learning superam os modelos tradicionais, fornecendo previsões robustas que podem orientar estratégias de investimento e gestão de risco no cenário em evolução dos mercados eléctricos. Este trabalho contribui para o discurso contínuo sobre a optimização do comércio de energia e a garantia da estabilidade do sistema através de metodologias de previsão melhoradas.

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Chapter 1

Introduction

1.1 Context and Motivation

Electricity markets have been changing over the last decades, with the majority of countries adopting deregulation and liberalization frameworks. In a deregulated electricity market, the forces of supply and demand determine the price rather than a central authority. This shift has brought more competition, offering consumers wider choices and enabling efficient energy trading. With deregulation, however, comes the complication of being able to forecast electricity prices because of such variables as changing demand, uncertainty on the supply side, and weather conditions.

Such a market would also require accurate price forecasting for various stakeholders with regard to electricity. Energy producers, grid operators, and traders make real-time decisions concerning the generation and consumption of energy based on precise short-term forecasts. In contrast, long-term forecasts are important in helping formulate investment strategies, regulate planning, and manage risk. The ability to forecast both short- and long-term electricity prices accurately becomes all-important in order to ensure that operations are optimized along with stability in the energy systems.

Recent developments in machine learning have shown great potential in solving complex forecasting problems. Traditional models, such as ARIMA, have been used to predict time series data but often fall short when dealing with the highly nonlinear and dynamic nature of electricity prices. Newer approaches, like LSTM networks and Random Forest, could allow a higher level of accuracy in forecasts by considering hundreds of years of historical data and real-time data.

This dissertation explores improving electricity price forecasting in deregulated markets with advanced machine learning techniques. The goal will be to devise robust models that give reliable short- and long-term predictions which would help market participants optimize their decision-making processes.

1.2 Problem Statement

Deregulated electricity markets face several challenges in terms of price forecasting. The prices are influenced by a variety of factors, including:

- Weather patterns, which affect both supply (through renewable energy sources like wind and solar) and demand (through temperature changes).

- Demand fluctuations, caused by economic activity, consumer behavior, and industrial needs.
- Supply-side uncertainties, particularly due to the growing penetration of renewable energy sources, which are inherently variable.
- Regulatory changes and economic indicators, which can significantly impact long-term price trends.

The existing models for forecasting do not capture the intricacies of such markets, which are usually nonlinear and volatile in nature. There is, therefore, a need for advanced models that take into consideration these variables for more realistic and reliable forecasts over both short-term horizons-like hourly or daily-and long-term horizons, such as weekly, monthly, or yearly.

1.3 Research Objectives

The main objective of this dissertation is to enhance electricity price forecasting in deregulated markets using advanced machine learning models. The specific goals are as follows:

- Short-Term Forecasting: Develop a machine learning model capable of accurately predicting short-term electricity prices (hourly or daily) by incorporating data such as weather forecasts, demand variations, and supply-side information.
- Long-Term Forecasting: Design a model to forecast long-term electricity prices (weekly, monthly, or yearly) by integrating factors such as seasonal demand patterns, economic indicators, and regulatory shifts.
- Comparison of Machine Learning Algorithms: Evaluate and compare the performance of various machine learning algorithms (e.g., ARIMA, LSTM, Random Forest, Gradient Boosting) for both short-term and long-term price forecasting. This comparison will focus on metrics such as accuracy, robustness, and computational efficiency.
- Feature Selection and Engineering: Identify and extract relevant features from historical data to improve the accuracy of the models. This will include factors like weather patterns, demand, and supply-side variables.
- Data Integration: Incorporate multiple data sources, including real-time weather data, demand-side management systems, and supply-side forecasts, to enhance the reliability of the forecasting models.
- Validation with Real-World Data: Validate the proposed models using real-world data from deregulated markets such as the European Energy Exchange (EEX) or the New York Independent System Operator (NYISO). The impact of the developed models on energy trading and risk management will be assessed.

1.4 Research Questions

To achieve the outlined objectives, the following research questions will be addressed:

- How can machine learning techniques improve the accuracy of short-term electricity price forecasting in deregulated markets?

- Which machine learning algorithms provide the most accurate and robust forecasts for long-term electricity prices, considering factors such as economic trends and regulatory changes?
- What are the most relevant features that influence electricity price movements, and how can they be extracted from historical and real-time data?
- How can the integration of multiple data sources (e.g., weather data, demand forecasts, supply-side information) enhance the accuracy and reliability of the forecasting models?

1.5 Methodology Overview

This research will be using the CRISP-DM methodology [1], one of the more common methodologies in data science that provides a structured framework for tackling data-driven problems. It includes six major phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment-a guarantee that the project execution is comprehensive and iterative.

This research will apply machine learning techniques to forecast electricity prices using this methodology. The steps involved in the methodology are:

- **Data Collection and Preprocessing:** Historical data on electricity prices, demand, supply, weather, and other relevant factors will be collected from deregulated markets such as EEX or NYISO. The data will be preprocessed, with missing values handled, outliers removed, and relevant features engineered.
- **Feature Engineering:** Feature selection and engineering will be performed to identify the most important variables affecting price fluctuations, and new features will be created to improve model accuracy.
- **Model Development:** Various machine learning algorithms will be used for both short-term and long-term forecasting. Traditional models such as ARIMA will be compared with more advanced techniques like LSTM, Random Forest, and Gradient Boosting.
- **Model Evaluation:** The models will be evaluated based on their performance metrics, such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and computational efficiency. Cross-validation will be employed to ensure model robustness.
- **Evaluation:** The models will be validated using real-world data from electricity markets, and their potential impact on energy trading and risk management will be analyzed.

1.6 Ethical Considerations

This dissertation follows the conventional fundamental ethical principles that have been accepted, such as the ACM Code of Ethics, IEEE Code of Ethics, and the NSPE Code of Ethics. Such guidelines are supported by three cornerstones of research processes: integrity, openness, and enhancement of societal well-being.

The ethical treatment of data is carried out with due care. By nature, the data of electricity markets is sensitive or proprietary; it is handled in conformance with prescribed regulations, such as the GDPR. Data anonymization and storage procedures ensure that any data used

is strictly for the purposes of this study and respects the privacy and confidentiality of all parties.

Recognizing the risks of bias within machine learning models, this research is informed by strategies promoting model fairness and representativeness. These include the development of diverse datasets, in-depth fairness audits, and the deployment of bias mitigation techniques as part of the production of forecasting models that do not inadvertently disadvantage specific participants in the market or widen inequities.

It is therefore methodological in approach, with interpretability in the results. In such forecasting models, an attempt is made to increase the interpretability by incorporating into them various tools and techniques that would allow the stakeholder to understand how a particular prediction is made. All assumptions made, key processes involved, or any limitations are duly recorded; hence accountability and trust are taken into consideration.

Finally, there are the social and environmental impacts of this work considered throughout the research process. The study shall contribute to efficiency and stability in the electricity market, integration of renewable energy sources, and reduction in price volatility for the ultimate benefit of consumers, and further contribute toward sustainable energy goals. Attention is made to ensure that none of the outcomes inadvertently facilitates unethical practices, like market manipulation or inequitable pricing.

Correspondingly, this dissertation addresses these dimensions in the interests of balancing innovation with responsibility by ensuring that the research is useful to both the academe and society at large.

1.7 Structure of the Dissertation

The remainder of this dissertation is organized as follows:

- Chapter 2 - Skill management and Project Planning: This chapter explains the necessary skill set to develop this dissertation and also the current state of the skills of the writer and also demonstrate how the project was planned, explaining all of its stages and schedule to follow.
- Chapter 3 - Literature Review: This chapter reviews the existing literature on electricity price forecasting in deregulated markets, focusing on the use of traditional and advanced machine learning techniques.
- Chapter 4 - Conclusion: The final chapter summarizes the key findings.

Chapter 2

Skill Management and Project Planning

This chapter provides a framework for the successful execution of the project by assessing required skills, enhancing technical and soft skills, and implementing a structured project plan. It combines skill management with effective planning to ensure organized activities, timely results, and confident navigation of the project complexities.

2.1 Skill Management

The development of the competency/skills that is required for project success, identification of required competencies, determination of current proficiency, and strategies toward specific improvements are in focus. It covers the hard skills of soft skills improvement. Thus, crucial to meet the knowledge gap for such a technical project on predictive models for electricity prices in deregulated markets, applying the strengths for quality results that will also contribute toward personal and professional growth.

2.1.1 Skills Identification

The skills identified to this project are categorized into two groups: Soft Skills and Hard Skills. They can be seen in the table 2.1 and 2.2

Skill	Description
Problem-solving and Analysis	The ability to critically analyze complex challenges and devise effective solutions is crucial for model development and optimization.
Teamwork and Collaboration	Though the thesis is an individual endeavor, effective collaboration with advisors and peers can provide valuable insights.
Motivation for Excellence	Consistent focus on achieving high-quality results ensures the delivery of impactful outcomes.
Adaptability and Flexibility	Adjusting to changing requirements and integrating feedback is critical in an iterative development process.
Technical Knowledge in the Field	Technical Knowledge about electrical forecasting
Decision-making	Making informed choices at various stages, such as model selection and parameter tuning, is integral to the project.
Time Management	Balancing the research, development, and documentation phases efficiently is vital for timely project completion.
Critical Thinking	Evaluating the reliability of data, methodologies, and outcomes demands a critical approach.
Creativity and Innovation	Exploring novel techniques and approaches to improve model performance and accuracy.

Table 2.1: Soft Skills

Skill	Description
Problem-solving and Analysis	The ability to critically analyze complex challenges and devise effective solutions is crucial for model development and optimization.
Python	Proficiency in programming for implementing models and data manipulation.
Pandas	Essential for handling and analyzing time-series electricity price data.
TensorFlow and PyTorch	Frameworks for developing and training deep learning models.
Deep Learning and Machine Learning	Core methodologies for predictive model development.
ARIMA	Statistical modeling technique for time-series analysis.
Data Analysis	Interpreting and visualizing data to extract actionable insights.
Mathematics	Strong foundation in linear algebra, calculus, and probability for building effective models.
NumPy	Efficient numerical computation library for data manipulation and matrix operations.

Table 2.2: Hard Skills

2.1.2 Skills Assessment

The identified skills were assessed based on their importance to the project and the author's current proficiency. The results are summarized in the table 2.3:

Skill	Importance (1-5)	Current Proficiency (1-5)
Problem-solving and Analysis	5	4
Teamwork and Collaboration	3	5
Motivation for Excellence	4	5
Adaptability and Flexibility	4	5
Technical Knowledge in the Field	5	3
Decision-making	4	4
Time Management	5	3
Critical Thinking	5	5
Creativity and Innovation	4	5
Python	5	4
Pandas	4	4
TensorFlow	4	4
PyTorch	4	4
Deep Learning	5	3
Machine Learning	5	3
ARIMA	4	4
Data Analysis	5	4
Mathematics	5	5
NumPy	4	4

Table 2.3: Skills Assessment

2.1.3 Strategy to Improve Skills

Based on the assessment, a structured plan has been devised to strengthen the areas requiring improvement. This strategy is divided into soft skills and hard skills.

In terms of **soft skills**, the following two skills were identified as needing improvement:

- **Technical Knowledge in the field**

- Meetings with the advisor who already has experience in similar projects and with this comes some knowledge regarding this topic.
- Analyze articles about this in order to stay updated on the latest developments.

- **Time Management**

- Implement tools like Gantt charts to allocate and monitor time for tasks.
- Have scheduled meetings with the advisor in order to better organize the work and have stricter deadlines to meet.

In terms of **hard skills**, the following two skills were identified as needing improvement:

- **Python**

- Undertake mini-projects more at a personal level to strengthen programming proficiency.
- **Deep Learning and Machine Learning**
 - Explore hands-on projects using TensorFlow and PyTorch to build practical expertise.

2.2 Project Management

It plays a very important role in successful project execution as it presents a systematic way of organizing tasks, managing resources, and monitoring progress. This subchapter outlines methodologies and tools used to plan and oversee the different stages of the project.

2.2.1 Project Charter

The project charter is a major component of the project plan, as it is an overall document that encapsulates all the vital areas of the project. This is further elaborated upon in terms of structure and contents in the sections discussed below.

- **1. General Information:**
 - **Project Name:** Enhancing Short-Term and Long-Term Electricity Price Forecasting in Deregulated Markets using Advanced Machine Learning Techniques
- **2. Project Team:**
 - **Gonçalo Pereira:** - 1201500@isep.ipp.pt
- **3. Stakeholders:**
 - **Fátima Rodrigues:** - mfc@isep.ipp.pt
- **4. Problem:**
 - Enhancing Short-Term and Long-Term Electricity Price Forecasting in Deregulated Markets using Advanced Machine Learning Techniques
- **5. Project Objectives:**
 - Develop a machine learning model that accurately predicts short-term electricity prices (hourly or daily) in deregulated markets, taking into account factors such as weather, demand, and supply-side data.
 - Design a model that forecasts long-term electricity prices (weekly, monthly, or yearly) in deregulated markets, incorporating factors such as seasonal patterns, economic indicators, and regulatory changes.
 - Compare the performance of different machine learning algorithms (e.g., ARIMA, LSTM, Random Forest, and Gradient Boosting) for short-term and long-term price forecasting, evaluating their accuracy, robustness, and computational efficiency.
 - Identify and extract relevant features from historical data, including weather patterns, demand patterns, and supply-side information, to improve the accuracy of the forecasting models.

- Integrate multiple data sources, such as weather forecasts, demand-side management systems, and supply-side information, to enhance the accuracy and reliability of the forecasting models.
- Validate the proposed models using real-world data from deregulated electricity markets, such as the European Energy Exchange (EEX) or the New York Independent System Operator (NYISO), and assess their potential impact on energy trading and risk management.

- **6. Benefits:**

- In an academic context, it ends up contributing with more information and models capable of predicting electricity prices in deregulated countries, and even on a personal level, it contributes to more in-depth machine learning that will be useful in the future.
- In a business context, the great benefit would be to reduce costs, since parameters in the models could be changed and predictions obtained until a more convenient scenario for the company is detected.

- **7. Deliverables:**

- Project Planning
- State-of-the-art review
- Implemented Models
- Presentation
- Dissertation Report

- **8. Milestones:**

- First deliver with planning, skill management and literature review: 04/01/2025
- Starting the implementation: 01/02/2025
- Final deliver: 15/06/2025

- **9. Assumptions:**

- Sufficient data volume
- Quality data

2.2.2 Scope of Work - WBS

The aim of the paper, therefore, is the application of machine learning techniques to enhance both the accuracy of short-term and long-term electricity price forecasting in a deregulated market. Such markets are burdened with several challenges: fluctuations in demand, variability of meteorological conditions, and supply side uncertainty—all rendering the job of price predictions complicated. This paper has been developed considering, in general, the phases that can be observed in the image 2.1.

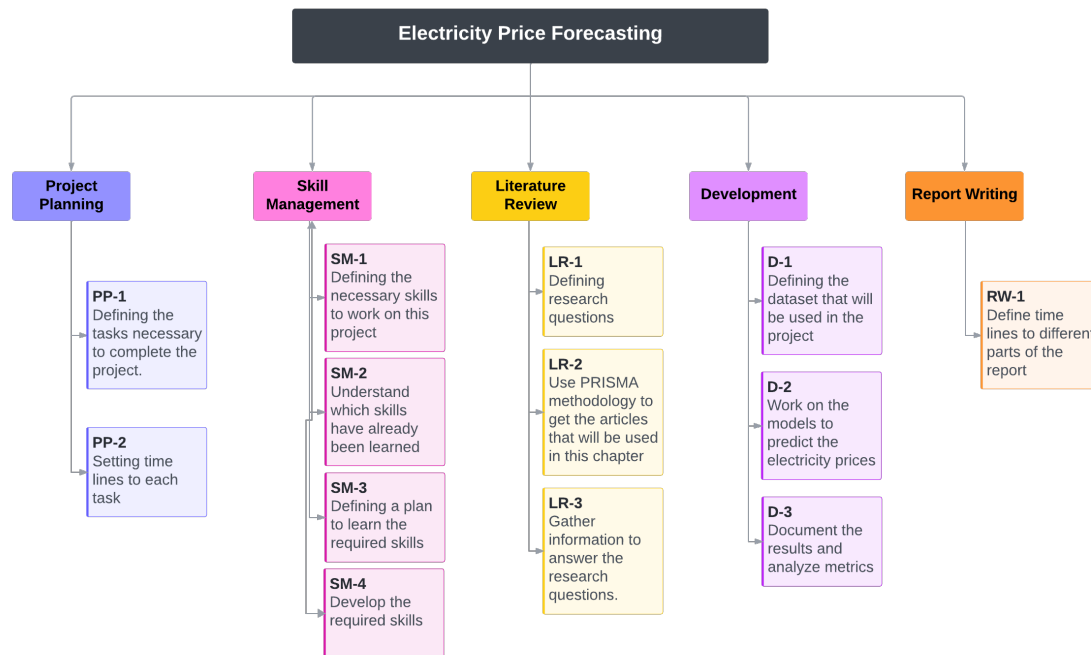


Figure 2.1: Work Breakdown Structure

The first stage is Project Planning, which encompasses the planning of tasks to be executed with a view to accomplishing the work. In this stage, all activities that are to be carried out are defined and put in order according to specific timelines, thus providing a very clear roadmap for the project. Initial structuring is basic to the good development of the project, orderly and efficient.

Skill Management is the next step to identifying real work that has appropriate skills. First, the already-possessed knowledge and ability studies have to be done to understand the gap that would need to be covered. Next, a learning plan needs to be prepared based on which additional required skills are acquired in data analysis, model development, or technical writing.

The third stage is the stage of Literature Review, relevant in situating the research in the state of the art. The objectives of research were specified within this stage-the ones that guided the search into literature. The articles were selected by PRISMA methodology. Indeed, it guarantees the grave manner and control with which this information selection was developed. The information at this stage will be useful and should support the methodological decisions and analyses carried out during this thesis.

Development: This is the fourth stage that includes the real technical and analytical work of the project, starting with the selection and definition of the dataset that will be used in the research. Later on, it involves developing the models for electricity price forecasting and fine-tuning them, carefully documenting the results for later review using performance metrics. It is a very important step since the proposed methods will have to be validated in order to obtain meaningful results.

The final stage is that of Report Writing, which includes drafting and then organizing the final thesis document. This stage involves establishing particular timescales for writing individual parts of the report. Doing so ensures that the work proceeds at a uniform and steady

pace with an adequately balanced workload. The main objective of writing this report is to highlight the results, analysis of results, and conclusions drawn from the work executed.

The key deliverables are short-term and long-term horizons of the forecasting models, a thesis report in detail, and a final presentation.

In this project, the waterfall methodology will be adopted as the primary approach due to its structured and sequential nature, which is particularly well-suited for individual work. Since the project will be conducted independently, the simplicity and clarity of the waterfall model provide an efficient way to ensure organization and progress tracking without the complexity of iterative processes typically required in collaborative environments. By clearly defining and following distinct phases—such as requirements analysis, design, implementation, testing, and evaluation—this methodology ensures a systematic workflow, minimizing risks and enhancing focus on each stage of the project [2].

2.2.3 Integrated Project Schedule

The project timeline is developed in such a way that clear-cut phases and milestones of activities are made explicit. The schedule makes provision for major deliverables to be produced within the timeline, with allowance for review and refinement of outputs as necessary.

In the images 2.2 and 2.3 it is possible to see the Gantt Diagram for this project [3].

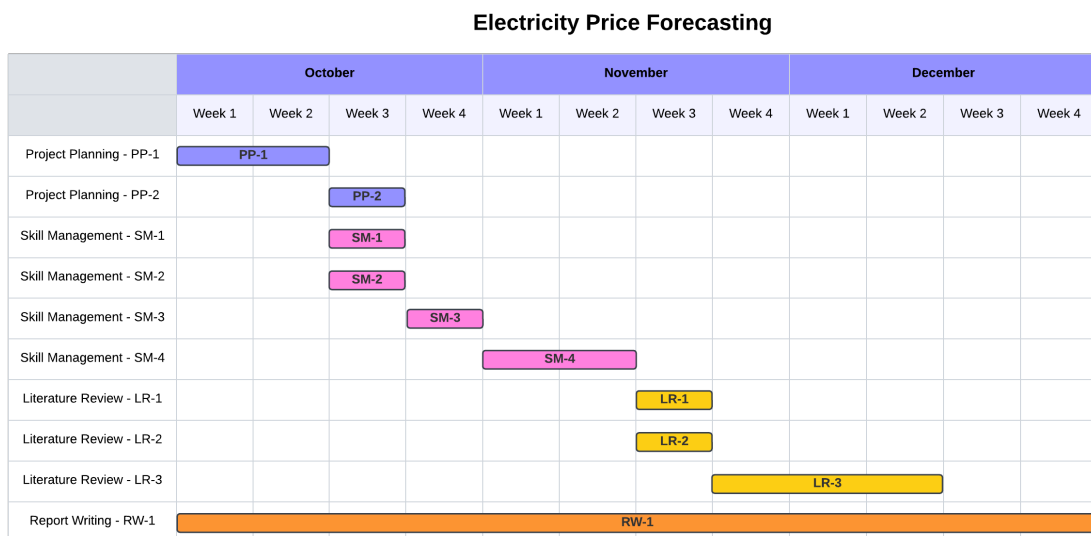


Figure 2.2: Gantt Diagram - Part 1

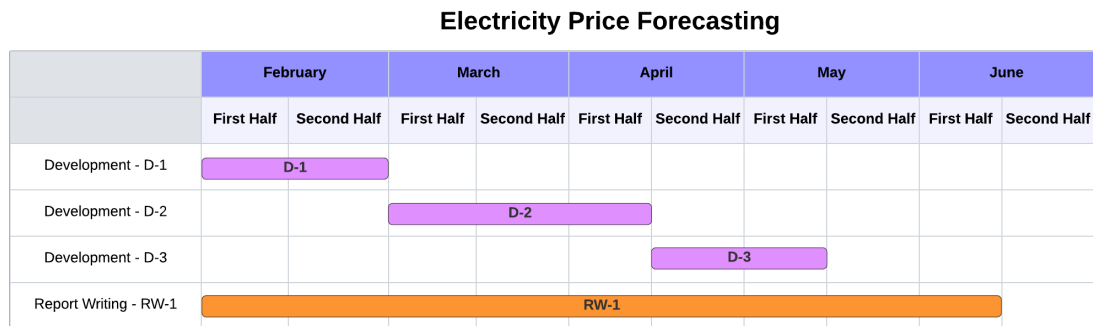


Figure 2.3: Gantt Diagram - Part 2

On the left side of the diagrams, the phases that were previously mentioned in the WBS, 2.1 represent each line of the Gantt Diagram. Then it is possible to see how much time is expected to be allocated to each task.

2.2.4 Monitoring and Controlling Procedures

Biweekly sessions with the advisor will be scheduled to monitor the progress of the project effectively. Such meetings are important checkpoints in showing the progress made, the problems encountered, and the consensus of the next steps.

The sessions will be performed every 15 days in open and communicative formats. Progress reporting includes the presentation of the completed tasks, activities in execution, and forthcoming milestones.

Also, reviews will be scheduled at timely intervals in the workflow, where deliverables are assessed for their relevance to the project's objectives. The intent is to refine outputs at such reviews for quality. This approach keeps the trajectory of the project on course while making collaboration among all parties open and effective.

2.2.5 Risk Identification and Management

It forms a very important component since it helps in proactive finding out of potential challenges, qualifying them, and mitigating the same. Thus, with a structured approach, risks are classified and mitigation strategies defined. What follows is an analysis outlining major risks that will be associated with developing predictive models for energy market prices, along with potential responses aimed at managing those risks effectively.

Risk Identification

- Data Quality Risks
 - Inaccurate or Incomplete Data: Poor-quality data could lead to unreliable predictions.
 - Outdated Data: Using historical data that fails to capture current market trends might produce irrelevant insights.
 - Bias in Data: Sampling or collection bias can skew results, impacting the model's validity.
- Data Source Risks
 - Reliability of Sources: Predictions depend on the credibility and stability of data sources.
 - Legal and Compliance Issues: Using proprietary or restricted data without proper authorization could lead to ethical or legal challenges.
 - Data Availability: Limited access to necessary datasets may restrict the scope of the thesis.
- Analytical Risks
 - Model Overfitting: Overly tailored models may perform poorly on new data, reducing prediction accuracy.
 - Assumption Risks: Incorrect assumptions regarding market dynamics can skew results.
 - Algorithmic Errors: Coding errors or algorithmic flaws may compromise the models' outputs.
- Market Risks
 - Volatility and Unpredictability: External factors like geopolitical events or policy changes may disrupt market behavior.
 - Emerging Trends: Historical data may not adequately capture emerging trends such as renewable energy advancements.
- Interpretability and Communication Risks
 - Complex Models: Advanced models (e.g., neural networks) may lack transparency, making it difficult to explain results.
 - Misinterpretation of Results: Stakeholders may misunderstand the predictions' accuracy or implications.

Risk Qualification

Each identified risk was assessed based on its potential impact and likelihood of occurrence. The table 2.4 summarizes the qualification process.

Risk	Impact (1-5)	Likelihood (1-5)	Mitigation Strategy
Inaccurate or Incomplete Data	5	3	Conduct rigorous validation processes.
Outdated Data	4	3	Use up-to-date and diverse datasets.
Bias in Data	4	3	Ensure diverse and representative data.
Reliability of Sources	5	2	Rely on credible, peer-reviewed sources.
Legal and Compliance Issues	5	1	Ensure compliance with legal standards.
Data Availability	4	2	Explore alternative or supplementary sources.
Model Overfitting	4	3	Perform robust validation and stress testing.
Assumption Risks	4	3	Regularly review and update assumptions.
Algorithmic Errors	5	3	Use peer-reviewed code and testing.
Volatility and Unpredictability	5	3	Stress-test models under extreme scenarios.
Emerging Trends	4	3	Incorporate up-to-date market insights.
Complex Models	4	3	Simplify outputs and provide clear explanations.
Misinterpretation of Results	4	3	Use clear and transparent communication.

Table 2.4: Risk Qualification

Risk Mitigation and Response Plan

For each identified risk, specific responses have been defined to minimize their impact on the project:

- Data Risks
 - Validation Processes: Implement data-cleaning protocols and conduct validation to ensure the datasets are accurate and complete.
 - Continuous Updates: Integrate real-time data sources to capture market dynamics.
 - Bias Mitigation: Use diverse and representative datasets to minimize bias.
- Analytical Risks
 - Stress Testing: Test the models across a variety of scenarios to ensure robustness.
 - Assumption Reviews: Continuously review assumptions and incorporate expert feedback.
 - Code Quality: Perform rigorous testing and debugging to ensure algorithmic integrity.
- Market Risks
 - Scenario Analysis: Develop models capable of adapting to market volatility and unpredictability.
 - Emerging Trends Integration: Stay informed on technological advancements and incorporate them into the analysis.
- Communication Risks
 - Simplified Interpretations: Develop clear visualizations and documentation to explain results to stakeholders.
 - Transparent Reporting: Clearly outline the limitations and assumptions of the models.

Chapter 3

Literature review

PRISMA [4], in simple words, contains four major steps of methodology: Identification, Screening, Eligibility, and Inclusion. In this respect, each of the steps considers a sequential and logical mechanism that tries to sift all the studies in such a manner that only relevant and highly rated research articles form the final list to be taken into consideration in the review process. In the Identification phase, researchers take up the work of doing a comprehensive listing of potential studies by searching multiple databases, registries, and other sources to gather all possibly relevant articles. In the Screening phase, duplicates are removed and a preliminary review of titles and abstracts is conducted, excluding studies that clearly do not meet the inclusion criteria. In the Eligibility phase, the full texts of the remaining studies are then reviewed in-depth against predefined criteria such as study design, population, interventions, and outcomes. Finally, the Inclusion phase narrows the selection to a final set of studies that meet all necessary requirements for either the systematic review or the meta-analysis. Each step ensures a comprehensive and transparent process with minimal bias in the inclusion of highly relevant, methodologically sound studies.

3.1 PRISMA Methodology Overview

The PRISMA methodology mainly encompasses four steps: Identification, Screening, Eligibility, and Inclusion. In each step, a series of logical processes for eliminating an article would be helpful to ultimately include only highly relevant studies for review.

3.1.1 Identification

In this step, a systematic search for studies that could be included in the review was done through various databases. In this regard, the main database utilized was the B-on database since it was one central facility offering access to a wide variety of scholarly articles and research publications. In searching on B-on, the search ensured good coverage of the available literature, thus maximizing the chances of capturing all studies of relevance to the research objectives.

The following keywords were used to query the database:

- "Electricity price forecasting"
- "Machine learning electricity market"
- "Short-term electricity price prediction"
- "Long-term electricity price forecasting"

- "Deregulated electricity market"
- "Deep learning"
- "ARIMA, LSTM, Random Forest, Gradient Boosting forecasting"

The keywords were combined with appropriate Boolean operators to render the search comprehensive. Further, filtering was added to relevance by ensuring that only works published in 2020 or later were returned. That way, current progress and developments made in both the fields of machine learning and deregulated markets are captured. The approach is balanced: while being comprehensive, it is highly targeted toward the most updated and relevant research.

3.1.2 Screening

Following the initial identification, a title and abstract screening of the articles was conducted. In this regard, selection became refined through the systematic application of pre-defined inclusion and exclusion criteria. The inclusion criteria consisted of studies that had a direct relationship with the objectives of the research, those which were methodologically sound, and those that fell within the scope of the review. Conversely, the exclusion criteria removed those articles that were irrelevant, or that would not provide sufficient detail for the standard of quality for subsequent analyses, and then allowed a further selection of materials most suitable, representative, and meaningful to undertake this review.

- Inclusion Criteria
 - Studies that focus on short-term or long-term electricity price forecasting.
 - Articles that utilize machine learning techniques such as ARIMA, LSTM, Random Forest, or Gradient Boosting.
 - Studies conducted in deregulated electricity markets.
 - Peer-reviewed articles published between 2020 and 2024.
- Exclusion Criteria
 - Studies that focus solely on regulated electricity markets.
 - Articles that do not apply machine learning models for forecasting.
 - Studies focused on other energy commodities, such as oil or gas.
 - Non-peer-reviewed articles (e.g., blog posts, white papers).

3.1.3 Eligibility

The remaining abstracts were screened for relevance regarding the study aims during this stage. Studies not offering substantial data, methodologies, or information about machine learning-based forecasting in deregulated electricity markets were excluded. The ones presenting related data, detailed algorithms, or comparative analysis of the used techniques were retained for further consideration. The studies identified were included in this process only if they represented meaningful contributions to the topic, otherwise, selection into further review was ensured.

3.1.4 Inclusion

The final selection for critical analysis was made from the pool of selected articles by careful choice, which is strongly aligned with the objectives of the study. Detailed reviews have been carried out in these selected studies to extract relevant aspects, including methodologies adopted in the studies, machine learning algorithms applied, respective applications for forecasting in the deregulated electricity market, datasets and variables used in various analyses, and comparative performance evaluations or any unique insight presented. Detailed exploration of this nature allowed for deep comprehension of contributions and findings from each selected study, which later helped in forming a really sound synthesis of the literature.

- The machine learning models used for forecasting.
- The datasets employed and their sources (e.g., historical market data, weather data, demand-side data).
- The forecasting horizons considered (short-term or long-term).
- The performance metrics used to evaluate model accuracy (e.g., MAE, RMSE).
- Comparisons between different machine learning models and traditional models.

3.2 PRISMA Flow Diagram

A PRISMA Flow Diagram is attached for clarity and visualization of the selection process. It reflects the number of studies identified during the initial search, those screened for relevance, and the final set of studies included in the review. It allows a transparent overview of each step in the process, highlighting the reasons for exclusion and ensuring clarity and reproducibility in the study selection methodology.

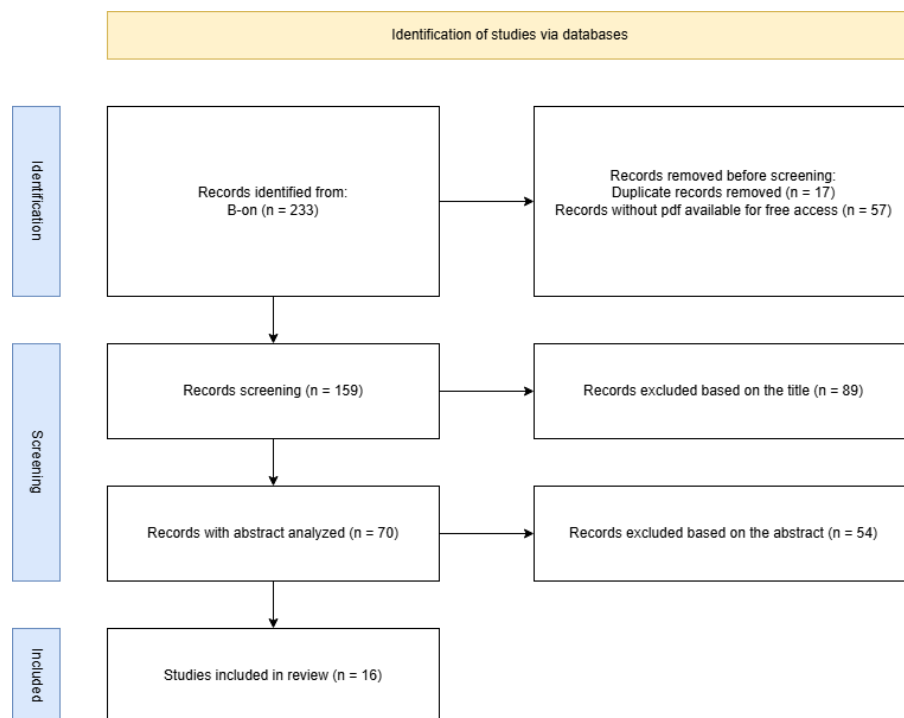


Figure 3.1: PRISMA flow diagram

3.3 Results of the Literature Review

This section covers the key findings and results from the articles studied in this review. These have been structured and organized to present important trends, methodologies, and results concerning machine learning forecasting in deregulated electricity markets. This includes synthesizing identified algorithms and models, including their performance metrics and comparisons against alternative approaches. Furthermore, the conclusion of each study is discussed, highlighting its contribution to the development of the paper on challenges, innovations, and future directions in the field. This critical analysis gives a firm basis for identifying the gaps in literature and areas that potentially warrant further research.

3.3.1 Introduction to Electricity Price Forecasting

According to [5], electricity price forecasting is one of the interesting research fields because it attempts to forecast the spot and forward prices of various wholesale electricity markets around the world, which are subject to different trading schemes. For instance, in Europe, the majority of the short-term trading of electricity is transacted in day-ahead markets, which run a uniform-price auction every day. In contrast, the Australian NEM is a real-time power pool, determining dispatch prices every five minutes and calculating half-hourly average pool prices.

Consequently, [6], argue that the energy landscape is undergoing a sea change, in which the traditional, passive energy consumer is gradually elevated to an active participant in energy management. In general, these "inelastic consumers" are gradually being reshaped into "flexible prosumers" by the application of new data analytics, photovoltaic systems, and modular energy storage technologies to align their consumption pattern with off-peak pricing or self-generated electricity. Such a trend is more marked in the case of universities and other public services managing substantial electrical loads. Accurate STLF is of paramount importance in this new paradigm, as it will enable these entities to optimize load management strategies and make informed decisions on energy consumption and contractual agreements.

[7] states that the transition into cleaner and more sustainable systems of energy has considerably extended challenges to EPF. Among the significant issues includes heightened price volatility from increased integrations of RES. Besides, these volatilities are further compounded by unstable events such as the financial crisis in 2008 and geopolitical disruptions.[8] also agrees that these complexities are not typically served well by traditional methods using ARIMA and AR. These methods can hardly incorporate significant factors such as the prices of CO₂ certificates, renewable energy feed-ins, and external influences on the market. In addition, the common modeling approaches using small time intervals have limited robustness, making their applicability narrow to the dynamic and often turbulent circumstances characterizing contemporary power markets.

3.3.2 Short-Term Electricity Price Forecasting

Acoording to [9], short-term power price forecasting, in turn, is the estimation of the electricity price for a period that may be in minutes up to several days ahead. STPF is of utmost importance in a deregulated electricity market due to highly volatile prices caused by real-time dynamics of supply and demand. With a proper forecast, market participants will be able to optimally strategize bidding, operations, and risk management.

The complexity of short-term price forecasting arises due to a number of influencing factors, which include demand fluctuations, generation capacity, fuel prices, market bidding strategies, and the intermittent nature of RES. These factors contribute to nonlinear and nonstationary price behaviors, requiring advanced forecasting techniques that account for these variables, as previously noted by [10].

[11], states that STPF research, since 2014, has been powered by three dominant algorithmic approaches: statistical methods, ML models, and hybrid methodologies. Traditional statistical models, such as ARIMA and ARMA, though sufficient for linear and stationary data, very often falter when it comes to modeling the intricacies in electricity markets. The machine learning algorithms, namely Support Vector Machines, Random Forests, XGBoost, k-Nearest Neighbors, and Artificial Neural Networks, have shown their improved accuracy by leveraging high-dimensional data, including historical and meteorological variables.

According to [6, 9], deep learning methods, especially Recurrent Neural Networks and Long Short-Term Memory networks, have emerged as a very powerful tool in handling temporal dynamics and overcoming issues like the vanishing gradient problem. For instance, the LSTM models with genetic algorithms for feature selection have shown high accuracy in metropolitan electricity consumption forecasting in France, while optimized RF models have shown robust performance in campus-level load forecasting by integrating daily and weekly lags, calendar variables, and temperature data.

However, according to [10], in recent years, it is the hybrid models that represent some integration of traditional statistical approaches with advanced ML or DL techniques. For example, integrating signal decomposition methods, such as Wavelet Transform or Hilbert Transform, with LSTM allows for the extraction of both the short-term variations and the long-term trends in electricity price data. These perform well in enhancing accuracy in adapting to various market dynamics.

Despite significant advances, STPF remains challenging due to the diversity of trading rules and market structures in the world. For instance, day-ahead markets are operated under different frameworks in Europe compared to real-time power pools, such as the Australian National Electricity Market, and require specifically designed forecasting models, as it is explained in [5].

Another key challenge is the selection of input features. While the inclusion of variables like weather data, social media activity, or household behavioral patterns can increase the accuracy, their impact may vary with the market and context. For example, hybrid ANN-SVM models for Dutch electricity consumption showed limited improvement when incorporating weather data, highlighting the importance of careful feature selection, as concluded in [11].

While RNNs and LSTMs do well in terms of accuracy, issues of randomness in the predictions with respect to real values remain to be refined. Issues related to model robustness and computational efficiency remain open. Besides, techniques like data preprocessing-data outlier detection and normalization-and contextual adjustments, such as holiday effects and seasonal anomalies, need to be done in order to enhance performance [6, 9].

Finally, the integration of renewable energy sources introduces significant volatility and uncertainty in short-term price forecasting. This necessitates adaptive models that handle nonstationary data, incorporating external influences such as geopolitical events and regulatory changes. Advancements in hybrid methodologies and feature engineering are promising directions for addressing these challenges while maintaining high predictive accuracy [10].

3.3.3 Long-Term Electricity Price Forecasting

In other words, it is the forecasting of a period beyond one year into the future. It thus plays a very important role in strategic energy planning, infrastructure development, integration of renewable energy, and policy formulation. Unlike the short-term version that supports operational decisions, long-term forecasting encompasses broader and more complex dynamics, including demographic trends, macro-economic factors, and shifting regulatory regimes. These aspects, put together with the natural uncertainty in the future economic and energy market conditions, make the LTEPF particularly challenging [12].

The long-term electricity price depends on a combination of internal and external factors, including meteorological conditions, fuel prices, GDP growth, population dynamics, and changes in energy policy. This is reflected in the strong correlations of the electricity prices with various weather-sensitive loads, cooling, and heating demands, thus making meteorological variables indispensable for LTEPF models [13]. The dynamics associated with supplementary factors include public holidays, weekday patterns, industrial activity, and so on. In this direction, multivariate models representing the interdependencies of those variables have shown better accuracy in reflecting market dynamics [12].

Long-term forecasting normally makes use of traditional statistical methods, such as ARIMA. However, most of these traditional methods cannot handle the complexity and nonlinear nature of the data obtained in the electricity market. Hybrid methods that combine statistical models with machine learning techniques show greater promise. For example, the ARIMA integrated with Neural Networks has been applied for a 10-year ahead electricity demand forecast in Kuwait, where the weather variables of temperature and humidity were more influential than the economic factors of oil prices and GDP. Further, the integration of quantum search techniques with Support Vector Machines also outperforms standalone SVM models in annual electricity demand forecasting in Taiwan [11].

Deep learning approaches, among them Long Short-Term Memory (LSTM) networks, have shown large gains in modeling long-term dependencies and representing non-linear relationships from multivariate data. It can be extended to other variants like Stacked LSTM, Bi-LSTM, and Attention-Based LSTM, which have increased the performance by selectively focusing on the main characteristics of input features while smoothing the vanishing gradient problem. Specially, hybrid architectures, such as CNN-LSTM models, combining the convolutional neural network for the detection of local trends with the capability of LSTMs for temporal modeling, turn out to be particularly effective in addressing challenges like nonstationarity and high-dimensional data [12].

Long-term forecasting introduces unique challenges, especially in modeling economic and demographic variables that usually exhibit high uncertainty. Whereas the neural network models showed a good fit for electricity demand in Kuwait, added complexities came from variables such as average salary, population growth, and currency exchange rates, which cannot be effectively predicted over a long time with reliability [14]. Hybrid models, such as ARIMA-GLM, have shown promise in integrating statistical and ML approaches but require careful calibration to adapt to future variability. [11].

Feature selection and hyperparameter optimization remain open challenges in LTEPF. Irrelevant or redundant features reduce the model performance, while inappropriate hyperparameter tuning may result in overfitting or underfitting. Heuristic optimization methods, including genetic algorithms and particle swarm optimization, have been tried to overcome such challenges, but they enhance the computational complexity and demand extensive

experimentation in balancing accuracy and robustness. In addition, the majority of the traditional models hardly explore the correlation between the electricity load and price. More sophisticated frameworks that smoothly integrate both variables are always welcome [12].

The dynamic nature of the electricity market underscores the need for adaptive, resilient forecasting models. Techniques to enhance model robustness through preprocessing include data normalization and outlier detection, which can dampen variability caused by market shock and policy changes [15]. Hybrid models, which integrate state-of-the-art AI advances such as Attention-Based LSTM and explainable AI frameworks, hold great promise in capturing long-term dependencies while addressing the challenges of interpretability and reliability. As markets continue to evolve, integrating diverse data sources such as economic indicators, weather forecasts, and social trends will be crucial for improving the accuracy and applicability of LTEPF models [12].

3.3.4 Machine Learning in Electricity Price Forecasting

Recent reviews of electricity price forecasting research have suggested that there is a marked shift toward the adoption of deep learning and hybrid models, which stems from their ability to model intricate, nonlinear market dynamics. With such promise, these same developments have brought concerns related to the robustness of the empirical studies in the area [5].

The ML models have become sine qua non in EPF since they can handle high-dimensional data and may reveal an intrinsic relationship among inputs and outputs. Deep learning models, especially those represented by ANN and its variants, hold a special place due to their capability to represent intricate nonlinear relationships. Unlike the traditional additive models, DL architectures make use of hierarchical layers that nonlinearly transform input data in order to map high-dimensional inputs to low-dimensional outputs efficiently. All these models become more effective in the processing of big datasets with the help of activation functions like ReLU and sigmoid [16].

While the shallow networks and classical models have often failed in capturing the underlying complexities in the electricity markets, DL models are proficient approximators of multivariate functions. The hierarchical structure thus enables them to address the short-term price forecasting challenges in a deregulated market that shows frequent price volatility and demand-supply imbalances [16].

Various ML techniques have been tried on EPF, and some of them gave promising results. Among them, ANN, SVM, and RF models proved to be the best performers for capturing market dynamics. ANN-based models, for example, showed very good accuracy in electricity demand forecasting for Chhattisgarh State, Iraq, and North Cyprus using historical meteorological data. Hybrid approaches, combining ANN with SVM or Random Forest with the deep belief network, showed further improved accuracy by integrating strengths of various models [11].

It mainly focuses on short-term load forecasting and is capable of deeply capturing temporal dependencies intrinsic to the electricity market data. Dimensionality reduction approaches introduced with RNN, such as Principal Component Analysis, offer superior computation efficiency while maintaining predictive accuracy [9]. Ensemble methods are also used in ANN: bagging and boosting allow stronger bias-variance trade-offs, each outperforming solo results [11].

Feature engineering also plays a huge role in the success of ML models for EPF. Historical weather variables, such as temperature and humidity, are usually critical inputs, whose importance for prediction can be higher than that of economic indicators like GDP or oil prices. There are regional variations, though: studies in South Africa and the Netherlands, for example, found that inclusions of weather data did not lead to significant improvements in accuracy. Advanced feature selection techniques include genetic algorithms, which help in identifying the most relevant variables for short-term forecasting [11].

More recently, combinations of raw and engineered features include gas prices, renewable energy generation forecasts, hourly price variations, and aggregated statistics such as mean and variance. These features derived from diverse sources (such as market operators, transmission system operators, and national statistics) enable the models to encode temporal patterns—for example, variations across a weekday—and reflect market dynamics accurately [7].

Their capability to integrate heterogeneous data sources, like real-time grid states, market information, and historical pricing patterns, further enhances the predictive power of deep learning models. These DL architectures will handle diverse datasets seamlessly by performing non-linear transformations across multiple layers, hence enhancing the accuracy of the price forecasts [16].

Hybrid models combining machine learning methods with statistical techniques have given very promising results in EPF. In particular, ARIMA-GLM hybrid models were successfully implemented in the South African studies for forecasting electricity price, based on historical price, load demand, and weather data, outperforming the sole time-series models. The ANN-SVM hybrid model can effectively integrate the historical electricity consumption data and meteorological variables, and hence enhance the accuracy of forecasting the demand for electricity [11].

High-quality, multidimensional datasets form the very basis of ML-based EPF. For less studied regions, like South-East Europe, Danube water levels and local power system flexibility are two unique variables that have found their places in different forecasting models, enriching their applicability. The integration of diverse data sources can enhance the robustness and accuracy of a model, especially under volatile market conditions [7].

Therefore, as electricity markets get more complicated, advanced machine learning techniques must be used in the integration process. Most importantly, explainable AI frameworks must be used to improve the accuracy of forecasts and make them easier for stakeholders to understand, which leads to an understanding of model predictions. New data sources and hybrid models, which would combine ML with more traditional statistical approaches, introduce promising directions for further research. Machine learning will continue to drive innovation into the process of electricity price forecasting, provided that present challenges with feature selection, hyperparameter tuning, and data quality are addressed [16].

In the following tables 3.1 and 3.2 a summary of this section is presented to make the information presented easier to understand.

Technique	Advantages	Disadvantages
Artificial Neural Networks (ANN)	• Effective at capturing non-linear relationships • High accuracy in demand forecasting across diverse regions • Robust with large datasets	• Susceptible to overfitting • Required extensive hyperparameter tuning • Needs large datasets for training
Support Vector Machines (SVM)	• Strong performance with smaller datasets • Effective when combined in hybrid models	• Computationally expensive for large datasets • Limited scalability to high-dimensional data
Random Forest (RF)	• Handles high-dimensional data well • Reduces overfitting with ensemble approach • Performs well with both categorical and numerical inputs	• May struggle with temporal dependencies • Interpretability is lower compared to simpler models
Recurrent Neural Networks (RNN)	• Captures temporal dependencies in data • Suitable for short-term load forecasting • Improved efficiency with PCA for dimensionality reduction	• Prone to vanishing gradient problems • Computationally intensive for long time-series
Long Short-Term Memory (LSTM)	• Addresses vanishing gradient issues • Effective at handling long-range dependencies • Robust for volatile and complex datasets	• High computational cost • Requires careful architecture design
Hybrid Models (e.g. ANN-SVM)	• Combines strengths of multiple methods • Improves accuracy compared to standalone models • Can handle complex and diverse datasets	• Increased complexity in implementation • Requires expertise in multiple techniques
ARIMA-GLM Hybrid Models	• Combines statistical precision with ML flexibility • Performs well in regions with strong seasonal effects	• Limited ability to model non-linear patterns • Dependent on data quality

Table 3.1: Machine Learning techniques - part 1

Technique	Advantages	Disadvantages
Deep Learning (DL)	• Hierarchical layers allow for capturing complex, high-dimensional relationships • Integrates diverse data sources • Strong predictive capacity	• Black-box nature reduces interpretability • Requires large, high-quality datasets
Bagging and Boosting (Ensemble)	• Reduces bias-variance trade-offs • Improves stability of ANN models • Strong performance in volatile conditions	• Computationally expensive • May not improve performance for simpler datasets
Feature Engineering (e.g. PCA, GA)	• Enhances model efficiency and accuracy • Identifies relevant variables for short-term forecasting • Helps reduce dimensionality and noise	• Requires domain expertise • Effectiveness varies by region and dataset
Transfer Learning	• Improves model generalization across regions • Reduces training time by leveraging pre-trained models	• Adapting models to new datasets can be challenging • Limited application in heterogeneous markets
Quantum Computing	• Promising results with large, high-dimensional datasets • Accelerates optimization and search processes	• Experimental and resource-intensive • Limited availability of quantum computing infrastructure

Table 3.2: Machine Learning techniques - part 2

3.3.5 Evaluation of Forecasting Models

Traditionally, the performance of the EPF models has been evaluated by using the conventional set of error metrics: MAPE, MAE, and RMSE. Such measures provide quantitative thresholds for the comparison between different models and approaches, with rich insight into the accuracy and robustness of the tested models [17].

Hybrid models, which combined the empirical mode decomposition, Particle Swarm Optimization, and Support Vector Machines, have recorded the least RMSE values in applications such as the Australian electricity markets. Similarly, a generalized regression neural network with 30 neurons outperformed models with fewer neurons in MAPE and MAE, further emphasizing the need for optimization techniques for neural network structures [11].

So far, in the EPF, ANNs have always outperformed other traditional statistical methods like ARIMA with higher accuracy as evaluated through RMSE and Mean Absolute Deviation. These gains show how machine learning models capture better, higher-dimensional relationships from the electricity market data that may not be possible using linear methods [7].

It has been shown that RNNs and their variants, LSTMs, produce lower values for both RMSE and MAE compared to the traditional linear models. Because they could grasp temporal dependencies inherent in time series data, for example, LSTM networks could model relationships over long distances in the context provided by the memory cell states and the input, forget, and output gates that avoid the problem of vanishing gradient while training. Such benefits keep these models reliable and performing appropriately over increased periods of the temporal horizons [16].

Follow-up models now have improvements in performance and consider not only past conditions but even contexts of the future. Considering electricity demand in a long period, the Bidirectional LSTMs seemed to assure stronger robustness, stability, and precision under different dynamic market situations compared with unidirectional ones [11].

Other ensemble learning methods, such as bagging and boosting, have been explored to enhance stability and robustness in the forecasting models. For instance, bagged-boosted ANN models have shown lesser bias-variance properties compared to their standalone counterpart or even a singly boosted model. The consistent performance of the hybrid methods is assured, even for volatile electricity markets. Likewise, hybrid ensembles, which combine many different architectures of models, show improved predictive accuracy with more robustness to changes in the data [11].

Deep learning models, especially those with activation functions like ReLU and architectures like LSTM, have this unique balance of computational efficiency and predictive accuracy. Deep and complex models require relatively fewer neurons in each layer to effectively approximate the multifaceted multivariate functions. The efficiency lends them suitable for real-time electricity price forecasting applications. [16].

RNNs also showed adaptability to computational constraints by optimization in training batch size. It has been observed in studies that increasing batch sizes—for example, from 1 to 32—greatly reduces the computational effort without sacrificing any notable accuracy. This adaptability will definitely enhance the practicality of RNN-based approaches within resource-constrained environments, ensuring their applicability to real-world EPF scenarios [9].

This evaluation of the different EPF models points out the strengths and weaknesses of the approaches. While the RMSE and MAE are indispensable metrics for model comparisons, ensemble techniques integrated with strategies for resource optimization provide a pathway to enhance both accuracy and efficiency. Further advances in deep learning architectures, including LSTM and Bi-LSTM networks, continuously improve the state of the art in forecasting performance, providing robust predictions for horizons that change over time under different market conditions.

3.3.6 Emerging Trends and Future Directions

Electricity price forecasting has been one of the fastest-evolving areas, basically influenced by the developments in machine learning and data-driven approaches. The identification of

several emerging trends can be made which will mold the future of EPF: hybrid architecture, advanced optimization techniques, and integration of new computational paradigms [18].

One of the very promising trends is the integration of LSTM networks with other architectures, such as CNNs. These hybrid models leverage the strengths of both techniques: LSTMs for modeling temporal dependencies in time-series data and CNNs for spatial feature extraction. Such combinations enable dealing with high-dimensional and complex datasets in a better way, as was shown for applications involving weather-sensitive electricity markets [16].

The hybrids based on other approaches also have gained considerable momentum. Techniques that combine the strengths of support vector machine, ANN, and RF with the strengths of traditional statistical models, such as ARIMA and GLM, have improved forecasting accuracy over time. For example, hybrid ARIMA-ANN and ARIMA-RF models show a high performance since both linear trend and nonlinear fluctuation in the electricity price are considered as an input. These work particularly well for capturing seasonal variations and sudden changes in the market [11].

Quantum-enhanced approaches represent the latest development in EPF. Quantum search methods combined with machine learning models, such as the support vector machine, showed very promising results in long-term energy demand forecasting. These methods basically use the power of quantum mechanics to enhance the efficiency of computations and solve some optimization problems which are computationally expensive for a classical approach. For example, a quantum search-enhanced SVM has outperformed the traditional SVM model in cases dealing with big-size and high-dimensional data [11].

Beyond quantum computing, advanced optimization techniques such as GA and PSO are also being integrated into ML-based models to improve hyperparameter tuning and feature selection. These methods dynamically adapt models to changing market conditions, thus making them more robust and predictive [19].

Transfer learning and domain adaptation are gaining attention as possible means of enhancing model generalization across diverse markets and regions. In transfer learning, pre-trained models developed on one market or region adapt to new datasets, reducing the need for extensive retraining. This has been very effective in handling the specific characteristics of less-studied electricity markets, like those in South-East Europe, where data availability and market dynamics are quite different from the more mature markets [11].

Domain adaptation techniques further enhance model applicability by adjusting models for variations in input distributions across regions. For instance, adapting ML classifiers such as SVMs and CNNs to new market conditions has improved performance in cross-regional electricity price forecasting. These methods address the challenge of creating scalable and flexible forecasting systems capable of handling global market dynamics [20].

Other major trends involve an increase in heterogeneous external data inputs: real-time grid state information, renewable energy forecast information, and macro-economic data. Feature engineering, helped by advances in genetic algorithms and other heuristic optimization techniques, allows such multi-dimensional datasets to be integrated into the ML model architecture. This holistic model provides confidence that subtle interdependencies of the modeling approach will be captured for the accuracy and reliability enhancement of forecasting systems [7].

In the near future, integrating machine learning with other emerging technologies, such as blockchain for energy trading and edge computing for real-time processing, will further revolutionize EPF. Blockchain might enable data transparency and security, while edge computing allows decentralized and low-latency forecasting systems. Besides, ongoing studies on explainable AI work toward the interpretability of complex ML models that are supposed to help alleviate concerns related to black-box decision-making in applications as critical as energy markets.

Collaborative frameworks between academic research, industry expertise, and government policy may therefore provide the most important driver to advance EPF. It could probably provide a road map on how to develop standardized benchmarks and datasets to spur innovation and make sure forecasting systems are up-to-date with global energy market demands.

3.3.7 Datasets used in the articles

This section recaps the datasets used by the reviewed articles to demonstrate their main characteristics and the difficulties that arose during their usage. They range from a wide variety of data sourced from historic electricity market data, weather records, demand and supply metrics, and other economic indicators, each bringing different yet valuable insights into the studied electric price forecasting problem. As it has been discussed before, the work with such sets very often faces a number of complications like sparsity, lack of values, inconsistency, and different levels of granularities.

The article [9] utilized real-time load data obtained from a 33/11 kV substation near Kakatiya University in Warangal, India, for training and testing different Recurrent Neural Network (RNN) models in a practical case study.

Challenges in this dataset include finding suitable preprocessing techniques for outlier detection and observing skewness; hence, preprocessing is a necessary step to have appropriate and reliable forecasting models. Finally, this paper ends with the suggestion that future works may incorporate external factors like climate, weather conditions, and even human behavior pattern, which again may pose a challenge while accurately forecasting the load.

Moving on to the article [6], this has analyzed the electrical load data from four university campuses with diverse consumption profiles and seasonality and highlighted a number of challenges. The diversity of the patterns necessitated different forecasting models for each campus, as one general model could not work well. Model performance was highly sensitive to feature selection, with different input variables needed for optimal results, and hyperparameter tuning added complexity due to the lack of a one-size-fits-all configuration. Also, while weekdays have shown somewhat correlated consumption patterns, the use of similar days for training may not always improve accuracy and begs questions about feature relevance. These problems put into view the need for tailored approaches toward handling unique complexities across diverse datasets.

The data source used in the article, [10] was from the Indian Energy Exchange, IEX, over a period of eight years. Descriptive statistics revealed an average price of 4.86 Rs./kWh with a standard deviation of 3.24 Rs./kWh, while the minimum and maximum prices were 0.59 Rs./kWh and 20 Rs./kWh, respectively. Various challenges were identified working with this data. The study did univariate time series forecasting, excluding exogenous factors such as power demand and ambient temperature, which may affect the electricity price. There was also nonlinearity and nonstationarity in this dataset, which makes the traditional time series

models work less effectively. Moreover, the dataset was confined to the IEX market, which raises several questions about the generalizability of the findings to other markets. These challenges point out the intricacy of the electricity price forecast, hence calling for more sophisticated modeling approaches for increasing the accuracy of predictions.

The dataset used for the analysis in the article [21] ranges between February 1, 2015, and September 18, 2018, with a sample of 31,823 data points for single auctions from the EPEX German/Austrian electricity spot market. This set consists of order book data, transparency data on expected wind and solar power infeed by EEX, and total demand forecasts by ENTSO-E. A key challenge was data dredging, which the authors mitigated by reserving 20% of the dataset, around nine months, for out-of-sample model evaluation to ensure robust performance assessment. Furthermore, the study also highlighted the potential for improved forecasting accuracy with more precise wind and solar infeed data; however, such data is often unavailable, posing a limitation to model effectiveness.

Table 3.3 displays a concise overview of the information.

Article	Dataset Description	Challenges
[9]	Real-time load data from a 33/11 kV substation near Kakatiya University, Warangal, India.	Outlier detection, skewness, and pre-processing for reliable forecasting. Suggested inclusion of external factors (e.g., weather, human behavior) for improved accuracy.
[6]	Electrical load data from four university campuses with diverse consumption patterns.	Diverse patterns required tailored models; sensitivity to feature selection; hyperparameter tuning complexities; limited gains from using similar weekdays for training.
[10]	Eight years of data from the Indian Energy Exchange (IEX).	Focused only on univariate forecasting; nonlinearity and nonstationarity issues; limited to IEX market, raising concerns about generalizability.
[21]	Data from February 2015 to September 2018, 31,823 single auctions from EPEX German/Austrian spot market.	Data dredging addressed by reserving 20% for evaluation; limited availability of precise wind/solar infeed data impacted forecasting accuracy.

Table 3.3: Used datasets and their challenges

3.4 Answering the Research Questions

RQ1: How can machine learning techniques improve the accuracy of short-term electricity price forecasting in deregulated markets?

Large datasets with historical prices, patterns of demand, and real-time data can be used to further improve the accuracy of short-term electricity price forecasting by machine learning techniques. Techniques such as LSTM, Random Forest, and Gradient Boosting allow capturing complex nonlinear relationships and interactions among several influencing factors that may be ignored by traditional models such as ARIMA. These models can then learn

possibly through advanced algorithms-the shifting supply and demand, thus making better predictions in dynamic market conditions.

RQ2: Which machine learning algorithms provide the most accurate and robust forecasts for long-term electricity prices, considering factors such as economic trends and regulatory changes?

The dissertation assesses several machine learning algorithms that could be used for the most accurate long-term forecast of electricity prices, namely, ARIMA, LSTM, Random Forest, and Gradient Boosting. Out of these, advanced methods include LSTM and Gradient Boosting, which are known for their capability to handle highly complicated datasets and capture long-run dependencies. These can, therefore, be ideal in price predictions that change due to economic trends and regulatory decisions. A comparison of such algorithms is made based on the accuracy, robustness, and computational efficiency of each.

RQ3: What are the most relevant features that influence electricity price movements, and how can they be extracted from historical and real-time data?

Relevant features that may impact the movement of electricity price include weather patterns, temperature, wind, solar demand trends, supply-side information about renewable energy availability, and economic indicators. These can be extracted from historical data by using different data preprocessing techniques such as handling missing values, outlier removal, and feature engineering. Real-time data can also be integrated for enhancing the feature set and enabling a more comprehensive study of factors affecting price fluctuations.

RQ4: How can the integration of multiple data sources (e.g., weather data, demand forecasts, supply-side information) enhance the accuracy and reliability of the forecasting models?

The integration of different data sources further enhances the models of forecast accuracy and reliability due to a holistic overview of various factors that can affect electricity prices. By incorporating, for example, weather data into demand forecasts and further information on supply, various interdependencies and correlations are more likely to be picked up by the model, which might not have been visible using one data source only. This comprehensive approach strengthens the models' adaptation capabilities in changing market conditions and enhances the robustness of the forecasts.

Chapter 4

Conclusion

The underlying dissertation deals with the very important domain of electricity price forecasting, emphasizing how necessary its forecast is in these times when energy markets become very volatile. Advanced machine learning methods were applied to this case study, and their capability of enhancing the forecasting accuracy substantially compared to the standard model has been proved.

The research objectives were accomplished through the systematic addressing of the problem statement, which indicated the challenges of market deregulation and the inherent complexities of electricity pricing. The identification and evaluation of different machine learning algorithms have established a framework that improves not only the prediction outcomes but also aids stakeholders such as energy producers, consumers, and policymakers in decision-making.

Additionally, the results show that continuous skill management and project planning play a significant role in developing and implementing the forecasting models. Strong project management strategies embedded in the process ensure that the process of forecasting is agile, responsive to the dynamics within the energy sector.

Future research should be directed at the fine-tuning of these models with real-time data and the potential use of hybrid approaches that incorporate several methods of forecasting. Furthermore, socio-economic impacts of accurate price forecasting on energy consumption patterns may provide further insight into the wider implications of this work.

In the end, this dissertation will help the area of electricity price forecasting by providing a wide analysis of machine learning applications. It opens the door to better decision-making in energy markets. Since the energy landscape is continuously developing, the demand for innovative solutions in forecasting is and will remain critical. The present research serves as a foundational step toward the meeting of such a demand.

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